FINAL RESCUE GUIDE

IGCSE Chemistry Complete Revision

Everything you need — atoms to organic chemistry — in 5 power-packed pages.
Rescue your grade today.

Cambridge 0620 | Atoms & Bonding | Stoichiometry | Reactions | Organics | Electrochemistry

⚛ ATOMIC STRUCTURE & PERIODIC TABLE

Subatomic Particles

Particle	Mass	Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	≈0	−1	Shells

- **Atomic number** = no. of protons
- **Mass number** = protons + neutrons
- **Isotopes** — same protons, different neutrons
- **Ions** — atoms that gain/lose electrons

Electron Shells & Configuration

- Shell 1 holds max **2** electrons
- Shell 2 holds max **8** electrons
- Shell 3 holds max **8** electrons (IGCSE level)
- Na = **2,8,1** · Cl = **2,8,7** · Ca = **2,8,8,2**
- Group number = **valence electrons**
- Period number = **number of shells**

Periodic Table Trends

- Across a period → atomic radius ↓, ionisation energy ↑
- Down a group → reactivity of metals ↑, non-metals ↓
- Group 1 = alkali metals · Group 7 = halogens · Group 0 = noble gases

CHEMICAL BONDING

Ionic Bonding

- Metal **loses** e^- → cation (+)
- Non-metal **gains** e^- → anion (−)
- Electrostatic attraction between ions
- High MP/BP, conducts when molten/dissolved
- Giant ionic lattice

Covalent Bonding

- Non-metals **share** electron pairs
- Simple molecular: low MP/BP
- Giant covalent (diamond, SiO_2): very high MP/BP, hard
- Does **not** conduct electricity
- H_2O, CO_2, CH_4, NH_3

Metallic Bonding

- Positive metal ions in a "sea" of delocalised electrons
- Good conductor of heat & electricity
- Malleable and ductile
- High MP/BP (strong bonds)
- Alloys — mix of metals, harder

⚖ STOICHIOMETRY & MOLES

$n = m / M_r$
moles = mass ÷ molar mass

$n = V / 24$
moles of gas (dm^3 at RTP)

$n = C \times V$
moles = conc × volume (dm^3)

Key Mole Concepts

- **1 mole** = 6.02×10^{23} particles (Avogadro)
- Molar mass = relative formula mass in g/mol
- At RTP: 1 mole of any gas = **24 dm^3**
- % purity = (pure mass / total mass) × 100
- % yield = (actual / theoretical) × 100
- Empirical formula = simplest whole number ratio
- Molecular formula = actual number of atoms

Balancing Equations — Steps

- Write word equation first
- Write correct chemical formulae
- Balance atoms — adjust coefficients only
- Check atoms on both sides match
- Add state symbols: (s), (l), (g), (aq)

Common Formulae to Know

H_2SO_4 HNO_3 HCl NaOH $Ca(OH)_2$ NH_3 Na_2CO_3
$CaCO_3$ $CuSO_4$ Fe_2O_3 Al_2O_3 MgO ZnO

ACIDS, BASES & SALTS

Acid–Base Reactions

- **Acid + Metal** → salt + hydrogen
- **Acid + Metal oxide** → salt + water
- **Acid + Metal hydroxide** → salt + water
- **Acid + Metal carbonate** → salt + water + CO_2
- **Acid + Ammonia** → ammonium salt

HCl + NaOH →	**NaCl + H_2O**
H_2SO_4 + $CaCO_3$ →	**$CaSO_4$ + H_2O + CO_2**

pH Scale & Indicators

Substance	pH	Colour (UI)
Strong acid	1–3	Red
Weak acid	4–6	Orange/Yellow
Neutral	7	Green
Weak alkali	8–10	Blue
Strong alkali	11–14	Purple/Violet

- Acid → H^+ ions in solution; Base → OH^- ions
- Neutralisation: $H^+ + OH^- \rightarrow H_2O$
- Strong acids fully dissociate (HCl, H_2SO_4, HNO_3)

⚠ **Exam Trap:** Salt name = metal from base + acid radical. HCl → chloride, H_2SO_4 → sulphate, HNO_3 → nitrate, H_3PO_4 → phosphate. NaOH + HCl → **Na**Cl (sodium chloride), not hydrogen chloride!

ENERGETICS

Exothermic vs Endothermic

	Exothermic	Endothermic
Energy	Released	Absorbed
ΔH	**Negative (−)**	**Positive (+)**
Surroundings	Get hotter	Get cooler
Examples	Combustion, neutralisation, respiration	Thermal decomposition, photosynthesis, dissolving NH_4NO_3

$$q = mc\Delta T$$
heat energy (J)

$$\Delta H = -q/n$$
enthalpy change

Bond Energies

- Breaking bonds = **endothermic** (energy IN)
- Forming bonds = **exothermic** (energy OUT)
- ΔH = bonds broken − bonds formed
- If ΔH negative → exothermic overall

⚡ REACTION KINETICS

Factors Affecting Rate

- **Temperature ↑** → particles have more energy → more frequent + energetic collisions → rate ↑
- **Concentration ↑** → more particles per volume → more collisions → rate ↑
- **Pressure ↑** (gases) → same as concentration ↑
- **Surface area ↑** → more particles exposed → rate ↑
- **Catalyst** → provides alternative pathway with lower activation energy → rate ↑

Collision Theory

- Reaction occurs when particles collide with **sufficient energy** (≥ E_a) and **correct orientation**
- **Activation energy (E_a)** = minimum energy needed
- Catalyst lowers E_a, more particles can react

⇌ EQUILIBRIUM

Le Chatelier's Principle

- If conditions change, equilibrium shifts to **oppose the change**
- **Temperature ↑** → shifts toward endothermic direction
- **Pressure ↑** → shifts toward side with fewer moles of gas
- **Concentration ↑** → shifts away from that substance
- Catalyst → **no effect** on position, only on speed

Haber Process	$N_2 + 3H_2 \rightleftharpoons 2NH_3$ (ΔH = −ve)
Contact Process	$2SO_2 + O_2 \rightleftharpoons 2SO_3$

Exam Tip: In the Haber process — high pressure (200 atm) favours NH_3 (fewer gas moles), low temperature favours NH_3 (exothermic) BUT a compromise ~450°C is used for a **reasonable rate**. Catalyst = iron.

ELECTROCHEMISTRY

Electrolysis Rules

- Requires molten or in solution
- (−) → cations reduced (gain e^-) → metals/H_2 deposited
- (+) → anions oxidised (lose e^-) → gases released
- In : H_2 at cathode, O_2 at anode
- In : H_2 at cathode, Cl_2 at anode
- In : Na at cathode, Cl_2 at anode
- In : Cu deposits at cathode, Cu dissolves from anode

OIL
Oxidation Is Loss (of e^-)

RIG
Reduction Is Gain (of e^-)

Oxidation Numbers — Rules

- Pure element = **0**
- H = **+1** (except in metal hydrides: −1)
- O = **−2** (except in peroxides: −1)
- Sum of oxidation numbers in compound = 0
- In ions = charge of the ion

REACTIVITY SERIES

Metals — Most → Least Reactive

Metal	Reaction with Water	Reaction with Acid
K, Na, Li	Vigorous with cold water	Explosive
Ca	Steady with cold water	Vigorous
Mg	Steam only	Vigorous
Al, Zn, Fe	Steam only	Moderate
Pb, Cu	No reaction	Slow/None
Ag, Au, Pt	No reaction	No reaction

Metal Extraction

- **Above carbon** (K→Al) → electrolysis required
- **Below carbon** (Zn→Pb) → reduction with carbon/CO
- **Below hydrogen** (Cu, Ag) → found native or reduction
- Blast furnace: $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
- Al extraction: electrolysis of molten Al_2O_3 (bauxite)

Rusting & Corrosion Prevention

- Rust needs **both O_2 and H_2O**
- $4Fe + 3O_2 + xH_2O \rightarrow 2Fe_2O_3{\cdot}xH_2O$
- Prevention: painting, oiling, galvanising (Zn coat), sacrificial anode
- **Galvanising** — physical + sacrificial (Zn more reactive than Fe)

⚠ **Remember:** **Displacement reactions** — a more reactive metal displaces a less reactive metal from its salt solution. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$ ✓ but $Cu + ZnSO_4 \rightarrow$ No reaction ×

ORGANIC CHEMISTRY

Alkanes (C_nH_{2n+2})

- Saturated hydrocarbons
- Single C–C bonds only
- CH_4, C_2H_6, C_3H_8, C_4H_{10}
- Combustion: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
- Substitution with Cl_2 (UV light)
- Cracking → alkenes + H_2

Alkenes (C_nH_{2n})

- Unsaturated — C=C double bond
- C_2H_4, C_3H_6, C_4H_8
- **Addition reactions**:
- + H_2 (hydrogenation) → alkane
- + Br_2 (bromine water) → decolourises → dibromoalkane
- + H_2O (steam, H_3PO_4) → alcohol
- Polymerisation → poly(ethene)

Alcohols (–OH group)

- $C_nH_{2n+1}OH$ functional group: –OH
- Methanol CH_3OH, Ethanol C_2H_5OH
- Combustion → $CO_2 + H_2O$
- Oxidation → carboxylic acid
- Fermentation: $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
- Dehydration (Al_2O_3) → alkene + H_2O

Carboxylic Acids & Esters

- Functional group:**–COOH**
- Ethanoic acid CH_3COOH (vinegar)
- Weak acid — partially dissociates
- Esterification: alcohol + acid → ester + water (H_2SO_4 catalyst)
- $CH_3COOH + C_2H_5OH \rightarrow CH_3COOC_2H_5 + H_2O$
- Esters have fruity smells, used in perfumes & flavourings

Polymers

- **Addition polymers**: alkenes join (no by-product)
- $nCH_2{=}CH_2 \rightarrow -(CH_2{-}CH_2)_n-$ poly(ethene)
- **Condensation polymers**: monomers with 2 functional groups → polymer + small molecule (H_2O)
- **Nylon**: diamine + dicarboxylic acid
- **Terylene (PET)**: diol + dicarboxylic acid
- Proteins: amino acids joined by peptide bonds (–CO–NH–)

KEY TESTS & EXPERIMENTS

Gas Tests

Gas	Test	Result
H_2	Lighted splint	Squeaky pop
O_2	Glowing splint	Relights
CO_2	Limewater	Turns milky
Cl_2	Damp litmus	Bleaches
NH_3	Damp red litmus	Turns blue

Flame Tests (metal ions)

Ion	Flame Colour
Li^+	**Crimson red**
Na^+	**Yellow/orange**
K^+	**Lilac/purple**
Ca^{2+}	**Brick red**
Cu^{2+}	**Blue-green**

Precipitate & Ion Tests

Test for	Reagent	Positive result
Cl^-	$AgNO_3$ (HNO_3)	White ppt
Br^-	$AgNO_3$	Cream ppt

I^-	$AgNO_3$	Yellow ppt
SO_4^{2-}	Ba^{2+} (HCl)	White ppt
CO_3^{2-}	Acid	CO_2 gas
NH_4^+	NaOH, warm	NH_3 gas

Final Exam Rescue Tips: Show ALL working in calculations · Always include units · State symbols matter · Draw dot-and-cross diagrams fully · In organic: name the type of reaction first · Check atom balance in every equation · Time management: 1 mark ≈ 1 minute